

Convolution & Z-Transforms

Revision notes with 10 fully solved questions

Scope: signal indexing, linear convolution, overlap methods, circular convolution, zero-padding, and introductory Z-transforms. The material follows Sections 1–6 of the supplied guide.

Conventions: finite sequences are zero outside their listed indices. Unlabelled lists start at $n = 0$. In a stem plot, horizontal position is the sample index and vertical height is the sample value. Here $u[0] = 1$.

1 Read a signal before calculating

A discrete-time signal $x[n]$ assigns a value to each integer index n . An arrow under a sequence entry marks $n = 0$. Always write the index row before calculating.

- $\delta[n - r]$ is 1 at $n = r$ and zero elsewhere.
- $u[n]$ is 1 for $n \geq 0$ and zero for $n < 0$.
- $u[n] - u[n - L]$ is a length- L rectangle: 1 at $n = 0, \dots, L - 1$.
- For $v[n] = x[an + b]$, a sample originally at m appears where $an + b = m$, if that gives an integer n ; its height stays unchanged.

Question 1 — Indexing, shifting and reflection

Given $x = \{1, 2, 2, 1\}_{n=-2, -1, 0, 1}$, find $x[-1]$, $x[3]$, $x[n + 1]$ and $x[1 - n]$. Also express $x[n]$ as shifted impulses.

Solution. The original samples are

$$\begin{array}{c|cccc} n & -2 & -1 & 0 & 1 \\ \hline x[n] & 1 & 2 & 2 & 1 \end{array}.$$

Thus $\boxed{x[-1] = 2, \quad x[3] = 0}$, since 3 lies outside the listed range.

For $v[n] = x[n + 1]$, solve $n + 1 = m$, giving $n = m - 1$. For $w[n] = x[1 - n]$, solve $1 - n = m$, giving $n = 1 - m$:

Original index m	Height $x[m]$	Index in $v[n]$	Index in $w[n]$
-2	1	-3	3
-1	2	-2	2
0	2	-1	1
1	1	0	0

Reading each result in increasing index order,

$$\boxed{x[n + 1] = \{1, 2, 2, 1\}_{n=-3, -2, -1, 0}}, \quad \boxed{x[1 - n] = \{1, 2, 2, 1\}_{n=0, 1, 2, 3}}.$$

Place an impulse at each original index and multiply it by that sample's height:

$$\boxed{x[n] = \delta[n + 2] + 2\delta[n + 1] + 2\delta[n] + \delta[n - 1].}$$

Check: substituting $n = -2, -1, 0, 1$ gives 1, 2, 2, 1, respectively. Every other index gives zero.

2 Linear convolution: flip, shift, multiply, add

For a linear time-invariant system with impulse response $h[n]$, each input sample produces a scaled, shifted copy of $h[n]$. Adding these copies gives

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k].$$

Here n is the output index and k is the summation index. Convolution is a sum of overlap products, not pointwise multiplication.

1. **Index:** if the input ranges are $a, \dots, b$ and $c, \dots, d$, the output lies within $a + c, \dots, b + d$.
2. **Flip:** change $h[k]$ to $h[-k]$.
3. **Shift:** form $h[n - k]$; a sample of h originally at m is now at $k = n - m$.
4. **Multiply and add:** multiply at matching k and sum. Repeat for each n .

For listed lengths L, M , retain $L + M - 1$ output positions, including any zero entries.

Question 2 — Linear convolution with negative indices

Find $y = x * h$ for

$$x = \{1, 2, 2, 1\}_{n=-2, -1, 0, 1}, \quad h = \{1, 3, 2\}_{n=-1, 0, 1}.$$

Solution. The output range is $-2 + (-1) = -3$ to $1 + 1 = 2$, with $4 + 3 - 1 = 6$ positions. The flipped sequence $h[-k]$ has heights 2, 3, 1 at $k = -1, 0, 1$.

Using the four possible nonzero samples of x ,

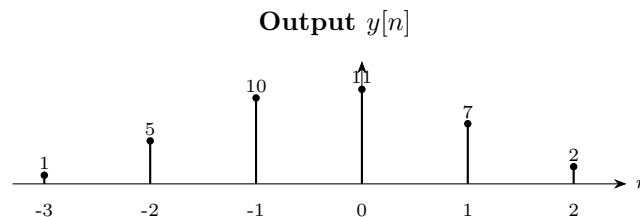
$$y[n] = h[n+2] + 2h[n+1] + 2h[n] + h[n-1].$$

Substitute each output index; only nonzero products are shown:

n	Overlap products	$y[n]$
-3	1(1)	1
-2	1(3) + 2(1)	5
-1	1(2) + 2(3) + 2(1)	10
0	2(2) + 2(3) + 1(1)	11
1	2(2) + 1(3)	7
2	1(2)	2

For instance, at $n = -2$, the shifted sequence $h[-2 - k]$ has heights 2, 3, 1 at $k = -3, -2, -1$. Only $k = -2, -1$ overlap x , so $y[-2] = 1(3) + 2(1) = 5$.

$$y = \{1, 5, 10, 11, 7, 2\}_{n=-3, -2, -1, 0, 1, 2}.$$



Check: $\sum_n y[n] = 1 + 5 + 10 + 11 + 7 + 2 = 36 = (1 + 2 + 2 + 1)(1 + 3 + 2)$. Outside $[-3, 2]$, there is no overlap and $y[n] = 0$.

3 Overlap counting and shifted rows

For two rectangles of unit height, every nonzero product is 1, so convolution counts overlapping samples. For general sequences, a shifted-row table organizes the same multiply–add calculation.

Question 3 — Convolution of two rectangular sequences

Find $y = x * h$ when $x = h = \{1, 1, 1\}$, both starting at $n = 0$.

Solution. Both inputs occupy 0, 1, 2, so y occupies 0, $\dots$, 4. The fixed sequence $x[k]$ is nonzero for $0 \leq k \leq 2$. The moving sequence $h[n - k]$ is nonzero when $0 \leq n - k \leq 2$, or $n - 2 \leq k \leq n$.

n	Nonzero k for $h[n - k]$	Overlapping k	Sum	$y[n]$
0	$-2, -1, 0$	$\{0\}$	1	1
1	$-1, 0, 1$	$\{0, 1\}$	$1 + 1$	2
2	$0, 1, 2$	$\{0, 1, 2\}$	$1 + 1 + 1$	3
3	$1, 2, 3$	$\{1, 2\}$	$1 + 1$	2
4	$2, 3, 4$	$\{2\}$	1	1

$$y = \{1, 2, 3, 2, 1\}_{n=0,1,2,3,4}.$$

Check: there are $3 + 3 - 1 = 5$ output positions and $1 + 2 + 3 + 2 + 1 = 9 = 3 \cdot 3$.

Question 4 — Signed samples using shifted rows

Find $\{2, -1, 1\} * \{1, 2\}$, with both lists starting at zero.

Solution. The output has $3 + 2 - 1 = 4$ positions. Since $x[0] = 2$, $x[1] = -1$ and $x[2] = 1$,

$$y[n] = 2h[n] - h[n - 1] + h[n - 2].$$

Write each scaled copy of h in its correct columns, then add:

Output index n	0	1	2	3
$2h[n]$	2	4	0	0
$-h[n - 1]$	0	-1	-2	0
$h[n - 2]$	0	0	1	2
$y[n]$	2	3	-1	2

Thus $y[0] = 2$, $y[1] = 4 - 1 = 3$, $y[2] = -2 + 1 = -1$ and $y[3] = 2$, giving

$$y = \{2, 3, -1, 2\}_{n=0,1,2,3}.$$

Check: $2 + 3 - 1 + 2 = 6 = (2 - 1 + 1)(1 + 2)$.

Useful properties and checks

$$x * h = h * x, \quad x * (h_1 + h_2) = x * h_1 + x * h_2, \quad x * \delta[n - r] = x[n - r].$$

Check the output length, the first and last products, and $\sum y = (\sum x)(\sum h)$. A correct total alone does not verify the order of the samples. Keep interior zeros and track the origin explicitly.

4 Circular convolution: rotating circles

An N -point circular convolution wraps all indices modulo N :

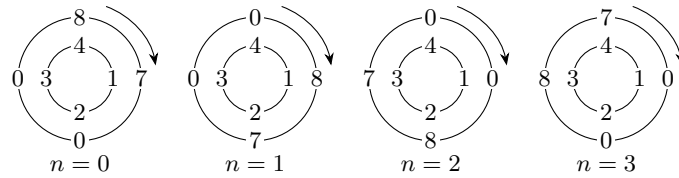
$$y_N[n] = \sum_{k=0}^{N-1} x[k]h[(n-k) \bmod N], \quad 0 \leq n < N.$$

Modulo N returns an index in $0, \dots, N-1$; for example, $(-1) \bmod 4 = 3$. The stored answer contains N samples; its periodic extension repeats these samples.

Question 5 — Four-point circular convolution with the ring method

Find the 4-point circular convolution of $x = \{1, 2, 3, 4\}$ and $h = \{7, 8\}$.

Solution. Pad the second input to $h = \{7, 8, 0, 0\}$. Place x clockwise on the inner ring, starting at the right. Place h anticlockwise on the outer ring. Read aligned products and add; then rotate the outer ring one position clockwise for the next output.



At fixed $k = 0, 1, 2, 3$ (right, down, left, up), the outer values and dot products are:

n	$(h[n], h[n-1], h[n-2], h[n-3])$	Dot product with $(1, 2, 3, 4)$
0	$(7, 0, 0, 8)$	$1(7) + 2(0) + 3(0) + 4(8) = 39$
1	$(8, 7, 0, 0)$	$1(8) + 2(7) + 3(0) + 4(0) = 22$
2	$(0, 8, 7, 0)$	$1(0) + 2(8) + 3(7) + 4(0) = 37$
3	$(0, 0, 8, 7)$	$1(0) + 2(0) + 3(8) + 4(7) = 52$

All h indices above are modulo 4. Each row is a right rotation of the preceding row.

$$y_4 = \{39, 22, 37, 52\}_{n=0,1,2,3}.$$

Check: $39+22+37+52 = 150 = (1+2+3+4)(7+8)$. The same result follows from $y_4[n] = 7x[n] + 8x[(n-1) \bmod 4]$.

Question 6 — Three-point circular convolution with a negative sample

Find the 3-point circular convolution of $x = \{1, 2, 0\}$ and $h = \{1, -1, 0\}$.

Solution. Circularly reverse h to get $(h[0], h[2], h[1]) = (1, 0, -1)$, then rotate right for each successive output:

n	Rotated h row	Dot product with $(1, 2, 0)$
0	$(1, 0, -1)$	$1(1) + 2(0) + 0(-1) = 1$
1	$(-1, 1, 0)$	$1(-1) + 2(1) + 0(0) = 1$
2	$(0, -1, 1)$	$1(0) + 2(-1) + 0(1) = -2$

$$y_3 = \{1, 1, -2\}_{n=0,1,2}.$$

Check: $1 + 1 - 2 = 0 = (1 + 2 + 0)(1 - 1 + 0)$.

5 Linear versus circular: choose the right length

	Linear convolution	N -point circular convolution
Index behaviour	Finite inputs are zero outside their ranges.	Input indices wrap modulo N .
Output length	$L + M - 1$ listed positions.	Exactly N stored positions.
Method	Flip, shift, multiply, add.	Reverse on a ring, rotate, multiply, add.
Tail	Remains at its own indices.	Folds onto indices with the same remainder modulo N .

Use the stated N and pad finite inputs with zeros as needed. If no N is given, $N = \max(L, M)$ is a common convention; state it explicitly. To reproduce the full linear result by circular convolution, the standard sufficient padding rule is

$$N \geq L + M - 1.$$

Repeating a sequence changes its samples; it is not zero-padding. An LCM may give a common period for periodic sequences, but is not the general padding rule for finite-list convolution.

Question 7 — Linear output, circular wrapping and zero-padding

For $x = \{1, 2, 3, 4\}$ and $h = \{7, 8\}$, find the linear convolution and obtain the 4-point and 5-point circular results from it.

Solution. The linear output has $4 + 2 - 1 = 5$ positions. Calculate every sample:

$$\begin{aligned} y_{\text{lin}}[0] &= 1(7) = 7, \\ y_{\text{lin}}[1] &= 1(8) + 2(7) = 22, \\ y_{\text{lin}}[2] &= 2(8) + 3(7) = 37, \\ y_{\text{lin}}[3] &= 3(8) + 4(7) = 52, \\ y_{\text{lin}}[4] &= 4(8) = 32. \end{aligned}$$

Hence $y_{\text{lin}} = \{7, 22, 37, 52, 32\}$ at $n = 0, \dots, 4$. For $N = 4$, index 4 wraps to index 0, so

$$y_4[0] = 7 + 32 = 39, \quad y_4 = \{39, 22, 37, 52\}.$$

For $N = 5$, pad both inputs to length 5. No nonzero tail wraps into the start, giving

$$y_5 = \{7, 22, 37, 52, 32\} = y_{\text{lin}}.$$

Check: both totals are $150 = (1 + 2 + 3 + 4)(7 + 8)$. The minimum length preserving all five linear positions is $N = 5$.

Question 8 — Selecting a length from the input sizes

Two finite lists have lengths $L = 4$ and $M = 6$. Find $\max(L, M)$, $\text{lcm}(L, M)$ and the minimum length guaranteed to reproduce their full linear convolution using circular convolution.

Solution. The larger input length is $\max(4, 6) = 6$. Since $4 = 2^2$ and $6 = 2 \cdot 3$, their least common multiple is $2^2 \cdot 3 = 12$. The linear output length is

$$N_{\text{min}} = L + M - 1 = 4 + 6 - 1 = 9.$$

At $N = 6$, possible linear samples at indices 6, 7, 8 fold onto 0, 1, 2, respectively. At $N = 9$, append five zeros to the length-4 input and three zeros to the length-6 input. All nine output positions remain distinct. At $N = 12$, the linear output is also reproduced, followed by three zeros, but the extra padding is unnecessary.

6 Z-transform: attach a power to each sample

The introductory Z-transform represents each sample as a coefficient and its index as a power:

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]z^{-n}.$$

Write the indices first, form $x[n]z^{-n}$ for every nonzero sample, and add. Negative indices give positive powers of z . For an infinite series, state its convergence condition. Only this basic convergence condition is used here.

Question 9 — Z-transforms of indexed sequences and their convolution

Find $X(z)$ and $H(z)$ for

$$x = \{1, 2, 2, 1\}_{n=-2, -1, 0, 1}, \quad h = \{1, 3, 2\}_{n=-1, 0, 1},$$

and multiply them to obtain the transform of their linear convolution.

Solution. Attach z^{-n} to each sample:

Index n	-2	-1	0	1
$x[n]$	1	2	2	1
$x[n]z^{-n}$	z^2	$2z$	2	z^{-1}

Therefore

$$X(z) = z^2 + 2z + 2 + z^{-1}, \quad H(z) = z + 3 + 2z^{-1}.$$

When multiplying transforms, exponents add. The coefficient at z^{-n} is $\sum_k x[k]h[n-k]$, so convolution corresponds to multiplication: $Y(z) = X(z)H(z)$.

$$\begin{aligned} Y(z) &= z^2(z + 3 + 2z^{-1}) + 2z(z + 3 + 2z^{-1}) \\ &\quad + 2(z + 3 + 2z^{-1}) + z^{-1}(z + 3 + 2z^{-1}) \\ &= (z^3 + 3z^2 + 2z) + (2z^2 + 6z + 4) \\ &\quad + (2z + 6 + 4z^{-1}) + (1 + 3z^{-1} + 2z^{-2}) \\ &= \boxed{z^3 + 5z^2 + 10z + 11 + 7z^{-1} + 2z^{-2}}. \end{aligned}$$

The coefficients at indices $-3, -2, -1, 0, 1, 2$ give $\{1, 5, 10, 11, 7, 2\}$, agreeing with Question 2. Each expression is finite for finite $z \neq 0$; $z = 0$ is excluded because negative powers occur.

Question 10 — Transform of a right-sided exponential

Find the Z-transform of $x[n] = (1/3)^n u[n]$ directly from the definition.

Solution. Since $u[n] = 0$ for $n < 0$ and $u[n] = 1$ for $n \geq 0$,

$$X(z) = \sum_{n=0}^{\infty} (1/3)^n z^{-n} = 1 + \frac{1}{3}z^{-1} + \frac{1}{9}z^{-2} + \frac{1}{27}z^{-3} + \dots$$

This is a geometric series with ratio $r = \frac{1}{3}z^{-1}$. Use $\sum_{n=0}^{\infty} r^n = 1/(1-r)$ when $|r| < 1$:

$$X(z) = \frac{1}{1 - \frac{1}{3}z^{-1}} = \frac{z}{z - \frac{1}{3}}, \quad |z| > \frac{1}{3}.$$

Check: the coefficients of $1, z^{-1}, z^{-2}, z^{-3}$ are $1, 1/3, 1/9, 1/27$, exactly the first four samples. More generally, $a^n u[n]$ has transform $1/(1 - az^{-1})$ for $|z| > |a|$, with $a \neq 0$.